+1 (412) 368-6051
metwaly.kareem@gmail.com
kareem-metwaly.github.io

Kareem Metwaly

github.com/kareem-metwaly
kareem-metwaly.github.io/scholar
linkedin.com/in/kareem-metwaly

Education

- **Ph.D.** in **Electrical Engineering**, Pennsylvania State University, USA. **August 2018 – December 2022**
  GPA: 3.96, **Thesis**: Explainable, Informed Deep Learning for Signal and Image Estimation.
- **M.Sc.** in **Engineering Mathematics**, Alexandria University, Egypt. **September 2014 – June 2018**
  GPA: 3.82, **Thesis**: Optimized Cooperative LTE/Wi-Fi Transmission over Unlicensed Bands.
- **B.Sc.** in **Electrical Engineering**, Alexandria University, Egypt. **September 2008 – June 2013**
  Grade: Distinction with Degree of Honor 88.72% (**GPA**≈3.94)

Work Experience

- **Machine Learning Engineer**, **Samsung Semiconductors,** San Diego, CA, USA **January 2023 – Present**
  - Developing and training deep learning models for computer vision.
  - Optimizing models for real-time inference on smart phones.
  - ML dataset collection, analysis, ML labeling and GenAI.
- **Machine Learning Research Engineer Intern**, **Scale AI**, San Francisco, CA, USA **June 2021 – December 2021**
  - Developing and implementing attributes prediction ML model and dataset labeling.
  - published a CVPR 2022 paper for the model and a dataset paper on arXiv.
- **Machine Learning Intern**, **Toyota Motor North America**, Mountain View, CA, USA **February 2021 – April 2021**
  - Developed an unsupervised GAN-based model for monocular depth estimation for night vision.
  - published two patents for training the style transfer and the depth estimation using style transfer.
- **Software Engineer** at **Brightskies Technologies**, Alexandria, Egypt **January 2017 – July 2018**
  - Developing and optimizing C/C++ algorithms on parallel CPU architectures for Seismic Imaging.
  - Presented four posters at multiple conferences.

Research/Teaching Experience

- **Teaching Assistant**, **Pennsylvania State University**, PA, USA **August 2022 – December 2022**
  Teaching and grading Linear Control Systems course in the Electrical Engineering Department.
- **Research Assistant**, **Pennsylvania State University** (iPAL lab), PA, USA **August 2018 – August 2022**
  Developing algorithms for Computer Vision and Inverse Problems in Image and Signal Processing:
  1. **Fast Deep Learning** model for HDR image generation (paper in ICASSP).
  2. **Deep Learning Non-Homogeneous Dehazing** (published paper and ranked **Second** in a **CVPR** Challenge).
  3. **Deep Learning System for Defect Detection and Classification** (funded by PPG, published a patent).
  4. **Deep Learning Algorithm Unrolling** for Radar (published a conference paper and a journal paper).
- **Mathematics Teaching Assistant**, **Alexandria University**, Alexandria, Egypt **April 2016 – June 2018**
  Teaching courses in Applied Mathematics for Engineering Students.
- **Research Assistant** at Collaborative Radio Cloud (CRC), **Alexandria University**. **May 2016 – August 2017**
  1. Participated in SC2 Challenge hosted by **DARPA**, USA.
  2. LTE/Wi-Fi Coexistence Project. [IEEE WCNC publication]
- **Research Assistant** for a collaboration between **E-JUST** and **Wasiela**, Egypt. **April 2014 – October 2015**
  Developing novel approaches for relaying using Network Coding. [IEEE ISCC publication]

Technical Skills

- **ML Frameworks and tools**: PyTorch, TensorFlow, Onnx, Quantization, Optuna, TensorBoard, MLFlow. **≈ 8 years**
- **Computer Vision:** Video Segmentation, Image Restoration, Classification, Object Detection, GenAI.
- **Programming Languages**: Python, C, C++, MATLAB, Linux – Bash. **> 10 years**
- **Tools:** Git, Docker, Jupyter, VSCode, Web Development, OpenMP, MPI, Valgrind, Intel Parallel Studio, Kubernetes, SQL.

Developed Toolboxes

- **CAR-API:** API to access my developed CAR Dataset with a PyTorch interface, github.com/kareem-metwaly/car-api
- **GlideNet:** Deep Learning model for multi-category attributes prediction, github.com/kareem-metwaly/glidenet
- **AtJwD:** Deep Learning for image dehazing (**second** in NTIRE 2020 competition), github.com/kareem-metwaly/AtJwD
- **Defect Detection and Classification:** Deep Learning for defect detection/classification [Proprietary of PPG Industries]

Publications, Posters and Patents

- **Kareem Metwaly**, et al. "**MIMO Radar Beampattern Design via Algorithm Unrolling**," IEEE Transactions on Aerospace and Electronic Systems, vol. 60, no. 6, pp. 9204-9220, Dec. 2024.
- **Kareem Metwaly**, et al. "**Interpretable, Unrolled Deep Radar Beampattern Design**," 2023 the IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), **2023**.
- **Kareem Metwaly**, Aerin Kim, Elliot Branson and Vishal Monga, "**GlideNet: Global, Local and Intrinsic based Dense Embedding NETwork for Multi-category Attributes Prediction**," 2022 the IEEE/CVF Computer Vision and Pattern Recognition (**CVPR**), **2022**.
- **Kareem Metwaly**, Aerin Kim, Elliot Branson and Vishal Monga, "**CAR – Cityscapes Attributes Recognition A Multi-category Attributes Dataset for Autonomous Vehicles**," to be submitted to NeurIPS 2023. arxiv: 2111.08243
- Li Yu*, **Kareem Metwaly***, James Wang and Vishal Monga, "**Surface Defect Detection and Evaluation for Marine Vessels using Multi-Stage Deep Learning**," the **IEEE transactions on Intelligent Transportation Systems** (TITS). *Equal Contrib. [under review]
- **Kareem Metwaly**, et al., "**NonLocal Channel Attention for NonHomogeneous Image Dehazing**," the IEEE/CVF Computer Vision and Pattern Recognition (**CVPR**) workshops, **2020**. "**Honorable Mention Award**" and "**Runner-up**" in **NTIRE 2020 competition**.
- Codruta O. Ancuti, Cosmin Ancuti, Florin-Alexandru Vasluianu, Radu Timofte, **Kareem Metwaly**, et al. "**NTIRE 2020 Challenge on NonHomogeneous Dehazing**," The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshops, **2020**.
- **Kareem Metwaly** and Vishal Monga, "**Attention-Mask Dense Merger (AttenDense) Deep HDR for Ghost Removal**," IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), Barcelona, Spain, **2020**, pp. 2623-2627.
- **Kareem M. Metwaly**, Karim G. Seddik, Mustafa Y. ElNainay, "**A Cooperative Scheme for the Coexistence of the LTE and WiFi Systems**" in Proceedings of the IEEE Wireless Communications and Networking Conference (**WCNC**), San Francisco, CA, **2017**.
- **Kareem Metwaly**, Mahmoud Alaa Eldin, Maha Elsabrouty, "**SISO and MIMO analog network coding relay architectures for the uplink of LTE-Advanced**" in proceedings of the IEEE Symposium on Computer and Communication (**ISCC**), Larnaca, Cyprus, **2015**.

Posters

- **Kareem Metwaly**, et al. from Brightskies Inc. and Philippe Thierry, Intel Corp, "**Accelerated Reverse Time Migration with optimized IO**" in HPC in Asia, ISC High Performance, Frankfurt, Germany, 2018.
- Essam Algizawy, Khaled Elamrawi, Kareem Morad, Ingy Monir, Mohamed ElBasyouni, **Kareem Metwaly**, Brightskies Inc. and Philippe Thierry, Intel Corp, "**Highly Scalable Flexi-trace-based Pre-stack Time Migration**" in HPC in Asia, ISC High Performance, Frankfurt, Germany, 2018.
- **Kareem Metwaly**, et al. from Brightskies Inc., Philippe Thierry, Intel Corp., "**Revisited RTM IOs strategies with new memory hierarchies**" in Intel eXtreme Performance User Group (IXPUG), KAUST, Thuwal, Kingdom of Saudi Arabia, 2018.
- **Kareem Metwaly**, Khaled Elamrawi, Essam Algizawy, Mohamed Mahmoud, Mohamed ElBasyouni, Brightskies Inc. and Philippe Thierry, Intel Corp, "**Accelerated Reverse Time Migration with optimized IO**" in Intel HPC Developer Conference, Denver, CO, 2017.

Patents

- **"Systems and methods for training a style model,"** US Patent 11,941,081 B2, Issued March 2024**.**
- **"Systems and methods for predicting depth using style transfer,"** US Patent 11,935,254 B2, Issued March 2024.
- **"System for asset management and coating maintenance,"** WO 2021/247961 A1, Published December 2021.

Honors and Awards

- "**Honorable Mention Award**" in the IEEE **CVPR** 2020 workshop and the **Second** in the **NTIRE** 2020 competition.
- Join RA/TA award from the Electrical Engineering Department at the Pennsylvania State University, PA, USA.
- Honored annually as a top student during my B.Sc. degree.

Talks and Conferences

- The IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), New Orleans, LA, USA. **June 2022**
- The IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) workshops, Seattle, WA. **June 2020**
- The IEEE Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), Barcelona, Spain. **May 2020**
- The IEEE Wireless Communications and Networking Conference (**WCNC**), San Francisco, CA, USA. **March 2017**
- The IEEE Symposium on Computers and Communications (**ISCC**), Larnaca, Cyprus. **July 2015**

Voluntary Activities

- Former **Web Manager** and former **Treasurer** of MGSO [a student organization at **Penn State**].
- Former **Activities Coordinator** of ISCP [a non-profit organization at **State College, PA**].
- Former volunteering teacher of technical courses for EE students [**IEEE Alexandria Student Branch**].
- Former volunteer in the **Library of Alexandria**, Egypt – 2 times winning of a world cultural competition [**UNESCO**].
- A former volunteer at a charitable non-profit organization named Sona' Elhaya [**Alexandria Egypt**].